RETAIL SALES DATA ANALYTICS AND AI-BASED SALES PREDICTION

A Python and Machine Learning Project Report

Submitted by: Ritwika Maji
Course / Internship: Data Analytics with AI
Institution: Bankura Unnayani Institute of Engineering
Submission Date: 12 October 2026

Note: This project uses a synthetic dataset created for educational demonstration. Confirm that this project meets the internship provider's submission rules before submitting.

Abstract

This project demonstrates a beginner-friendly data analytics workflow for retail sales. It inspects data quality, summarizes sales by product category, region and sales channel, visualizes patterns, and trains a Random Forest regression model to estimate sales amount. Python, pandas, Matplotlib and scikit-learn are used. The dataset is synthetic and does not represent a real business.

1. Problem Statement

Retail transaction data can help businesses understand performance and support planning. This project explores sales patterns across categories, regions and channels, and demonstrates whether transaction attributes can estimate sales amount in a sample dataset.

2. Objectives

• Inspect and summarize a structured dataset.

• Check missing values and calculate basic sales statistics.

• Compare sales by product category, region and sales channel.

• Create charts for easier interpretation.

• Train and evaluate a machine-learning regression model.

• Document limitations and responsible next steps.

3. Dataset Description

The CSV contains 600 synthetic records and eight source columns: Order_Date, Product_Category, Region, Sales_Channel, Units_Sold, Unit_Price, Discount_Percent and Sales_Amount. The data includes five categories, four regions and two sales channels. Month and day-of-week features are derived from the order date.

Measure	Value
Rows	600
Total sales	■11,582,841.27
Average sales per record	■19,304.74
Total units sold	2,686

4. Tools and Technologies

Python; pandas for data handling; NumPy for numerical operations; Matplotlib for visualizations; and scikit-learn for train/test splitting, Random Forest regression and model evaluation.

5. Methodology

Data inspection: Load the CSV, inspect rows and dimensions, and count missing values.

Feature preparation: Parse order dates, derive month and day of week, and one-hot encode categorical variables.

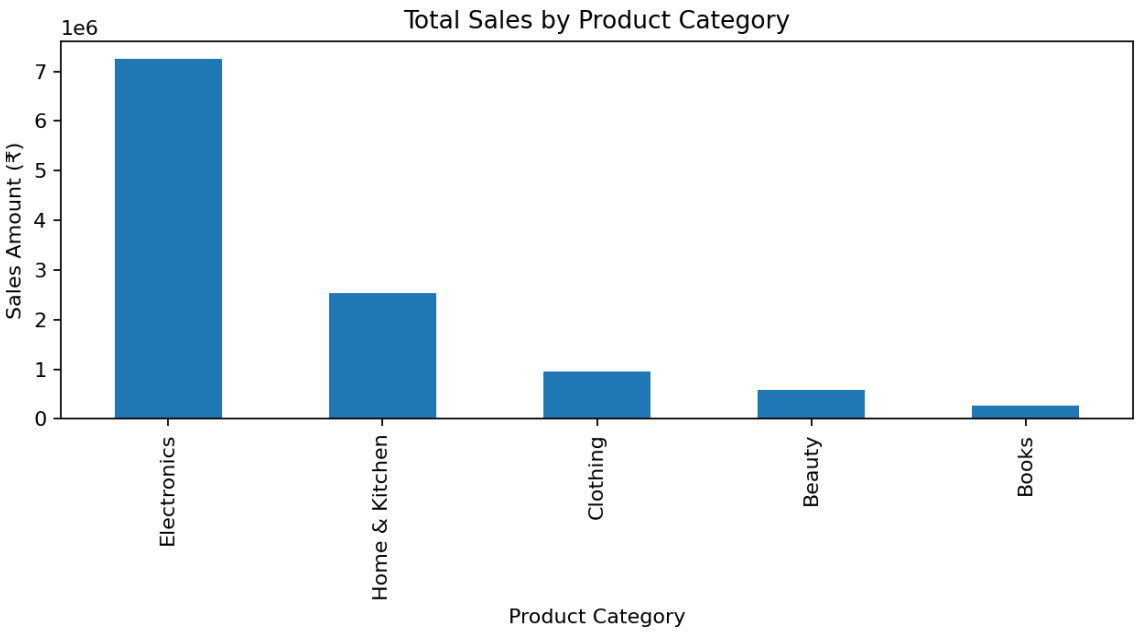
Exploratory analysis: Calculate sales totals by category, region and channel and plot the results.

Model training: Split records into 80% training and 20% testing data, then train a Random Forest Regressor.

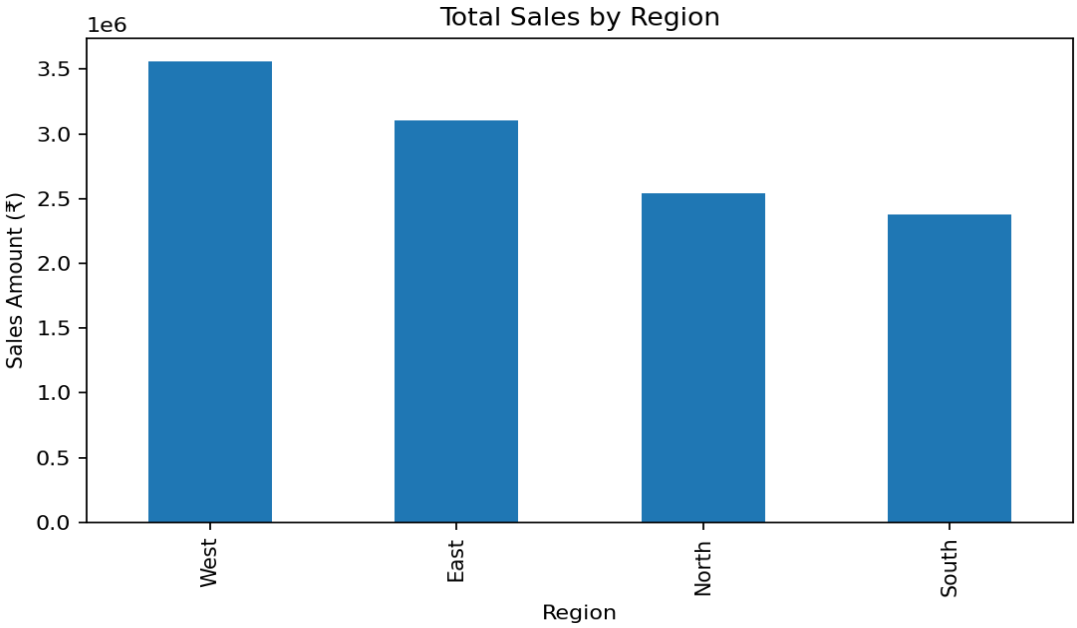
Evaluation: Evaluate predictions using MAE, RMSE and R^2 on the held-out test set.

6. Exploratory Data Analysis

Sales by product category:



Sales by region:

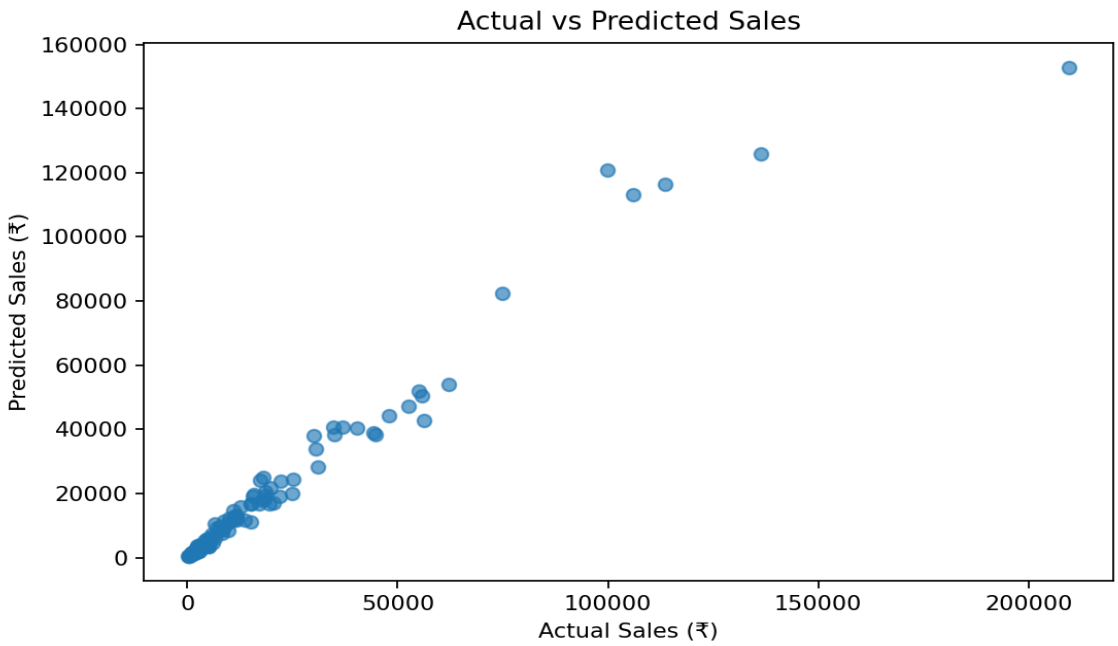


The charts summarize only this generated sample. The category and region rankings should not be generalized to real businesses.

7. Machine Learning Model

A Random Forest Regressor is used to estimate Sales_Amount from units sold, unit price, discount, calendar features, product category, region and sales channel. This model can capture non-linear patterns, but its usefulness depends on representative data.

Model evaluation values from the generated sample are printed when the Python script runs. They may vary if the dataset, library versions or settings change.



The plot compares actual and predicted sales for test records. Closer alignment indicates smaller prediction errors.

8. Findings and Recommendations

• Total sales in the sample are ■11,582,841.27 across 600 generated records.

• The highest sales category in this sample is Electronics; the highest sales region is West.

• The Online channel has the largest aggregate sales total in this sample.

• A dashboard could display total sales, average order value, category rankings, regional comparisons and monthly trends.

• For real use, validate data definitions, taxes, returns, outliers and date coverage with the business owner.

9. Limitations and Ethical Considerations

The dataset is synthetic, relatively small and based on simplified assumptions. It may not reflect real demand, stockouts, returns or market conditions. No personal information is included. Before using real data, obtain permission, remove personal identifiers and follow the provider's privacy rules. Predictions should not be the sole basis for important business decisions.

10. Conclusion

The project combines descriptive analytics and a basic machine-learning model in a reproducible Python workflow. It summarizes sales patterns and demonstrates sales amount estimation. Future work could use an approved real dataset, time-aware validation, stronger baseline comparisons and a dashboard.

11. How to Run

Install Python 3.10 or later. In the project folder run `pip install -r requirements.txt`, followed by `python retail_sales_analysis.py`. The script prints summaries and model metrics and saves charts in an outputs folder.

References

1. pandas documentation: https://pandas.pydata.org/docs/

2. scikit-learn documentation: https://scikit-learn.org/stable/

3. Matplotlib documentation: https://matplotlib.org/stable/

4. NumPy documentation: https://numpy.org/doc/